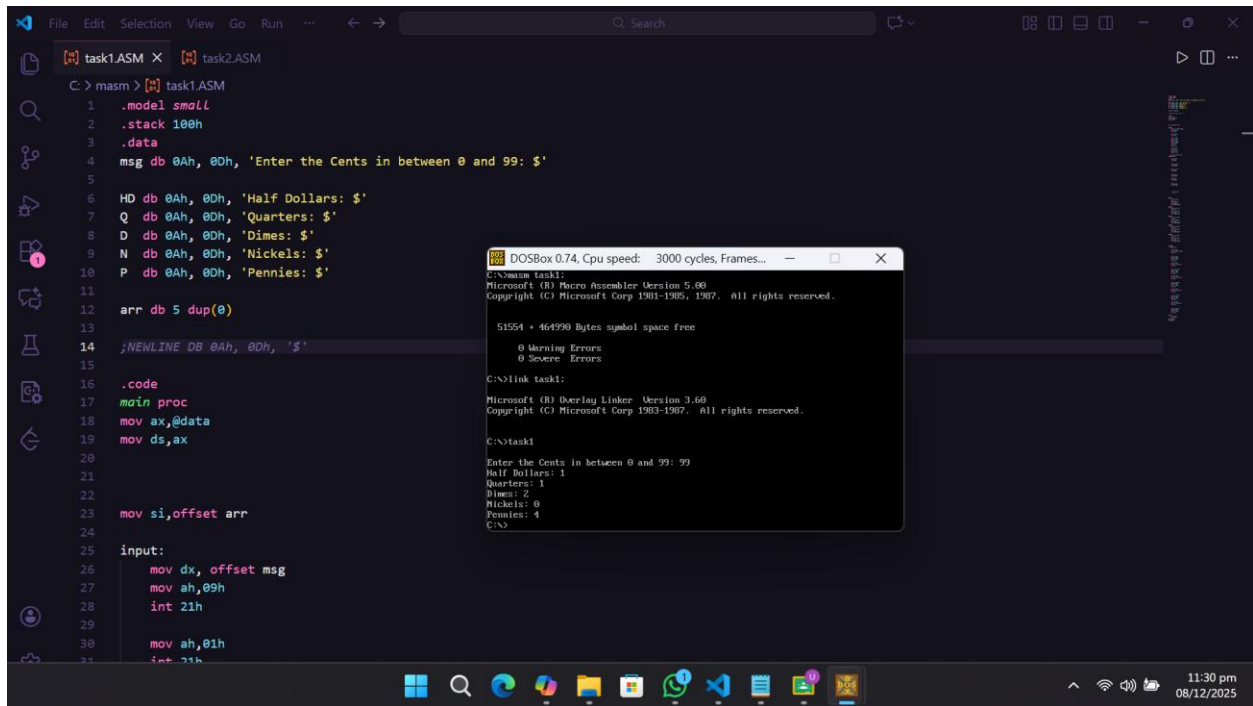


Task 1



The screenshot shows a Windows 10 desktop with a dark theme. In the background, a Visual Studio Code window is open with two files: `task1.ASM` and `task2.ASM`. The `task1.ASM` file is active, showing the following assembly code:

```
C:\> masm > task1.ASM
1  .model small
2  .stack 100h
3  .data
4  msg db 0Ah, 0Dh, 'Enter the Cents in between 0 and 99: $'
5
6  HD db 0Ah, 0Dh, 'Half Dollars: $'
7  Q  db 0Ah, 0Dh, 'Quarters: $'
8  D  db 0Ah, 0Dh, 'Dimes: $'
9  N  db 0Ah, 0Dh, 'Nickels: $'
10 P  db 0Ah, 0Dh, 'Pennies: $'
11
12
13 arr db 5 dup(0)
14 ;NEWLINE DB 0Ah, 0Dh, '$'
15
16 .code
17 main proc
18 mov ax,@data
19 mov ds,ax
20
21
22
23 mov si,offset arr
24
25 input:
26 mov dx, offset msg
27 mov ah,09h
28 int 21h
29
30 mov ah,01h
31 int 21h
```

In the foreground, a DOSBox 0.74 window is open, showing the output of the assembly and linking process. The window title is "DOSBox 0.74, Cpu speed: 3000 cycles, Frames...". The output text is as follows:

```
C:\>masm task1:
Microsoft (R) Macro Assembler Version 5.00
Copyright (C) Microsoft Corp 1981-1985, 1987. All rights reserved.

53554 + 464990 Bytes symbol space free

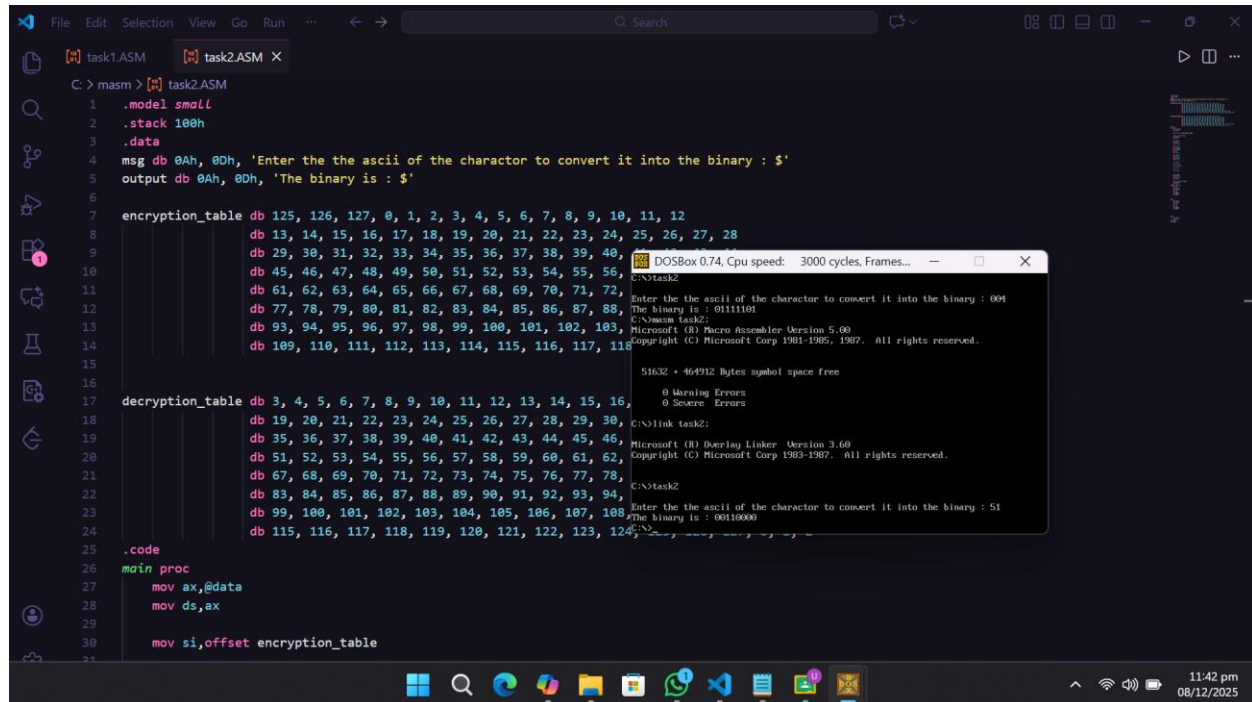
0 Warning Errors
0 Severe Errors

C:\>link task1:
Microsoft (R) Overlay Linker Version 3.60
Copyright (C) Microsoft Corp 1983-1987. All rights reserved.

C:\>task1
Enter the Cents in between 0 and 99: 99
Half Dollars: 1
Quarters: 1
Dimes: 2
Nickels: 0
Pennies: 4
C:\>
```

The Windows taskbar at the bottom shows the Start button, a search icon, and several application icons including Edge, File Explorer, and Visual Studio Code. The system clock in the bottom right corner indicates the time is 11:30 pm on 08/12/2025.

Task 1



The screenshot shows a Windows desktop environment. In the background, a Visual Studio Code editor is open with a file named `task2.ASM`. The assembly code is as follows:

```
C:\> masm > [M] task2.ASM
1  .model small
2  .stack 100h
3  .data
4  msg db 0Ah, 0Dh, 'Enter the the ascii of the character to convert it into the binary : $'
5  output db 0Ah, 0Dh, 'The binary is : $'
6
7  encryption_table db 125, 126, 127, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
8                    db 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28
9                    db 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40,
10                   db 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56,
11                   db 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72,
12                   db 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88,
13                   db 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103,
14                   db 109, 110, 111, 112, 113, 114, 115, 116, 117, 118
15
16  decryption_table db 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,
17                  db 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30,
18                  db 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46,
19                  db 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62,
20                  db 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
21                  db 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94,
22                  db 99, 100, 101, 102, 103, 104, 105, 106, 107, 108,
23                  db 115, 116, 117, 118, 119, 120, 121, 122, 123, 124
24
25  .code
26  main proc
27      mov ax,@data
28      mov ds,ax
29
30      mov si,offset encryption_table
31
```

In the foreground, a DOSBox window is running the program. It displays the following text:

```
C:\> task2
Enter the the ascii of the character to convert it into the binary : 004
The binary is : 01111101
C:\> task2
Enter the the ascii of the character to convert it into the binary : 51
The binary is : 00110000
```

The DOSBox window also shows the following status information:

```
DOSBox 0.74, Cpu speed: 3000 cycles, Frames...
C:\> task2
51632 + 464912 Bytes symbol space free
0 Warning Errors
0 Severe Errors
C:\> link task2:
Microsoft (R) Overlay Linker Version 3.60
Copyright (C) Microsoft Corp 1983-1987. All rights reserved.
```